These fields do not live *in* spacetime; they live, so to speak, one on top of the other: fields on fields. The space and time that we perceive in large scale are our blurred and approximate image of one of these quantum fields: the gravitational field.

Fields that live on themselves, without the need of a spacetime to serve as a substratum, as a support, and which are capable by themselves of generating spacetime, are called 'covariant quantum fields'. The substance of which the world is made has been radically simplified in recent years. The world, particles, light, energy, space and time – all of this is nothing but the manifestation of a single type of entity: covariant quantum fields.

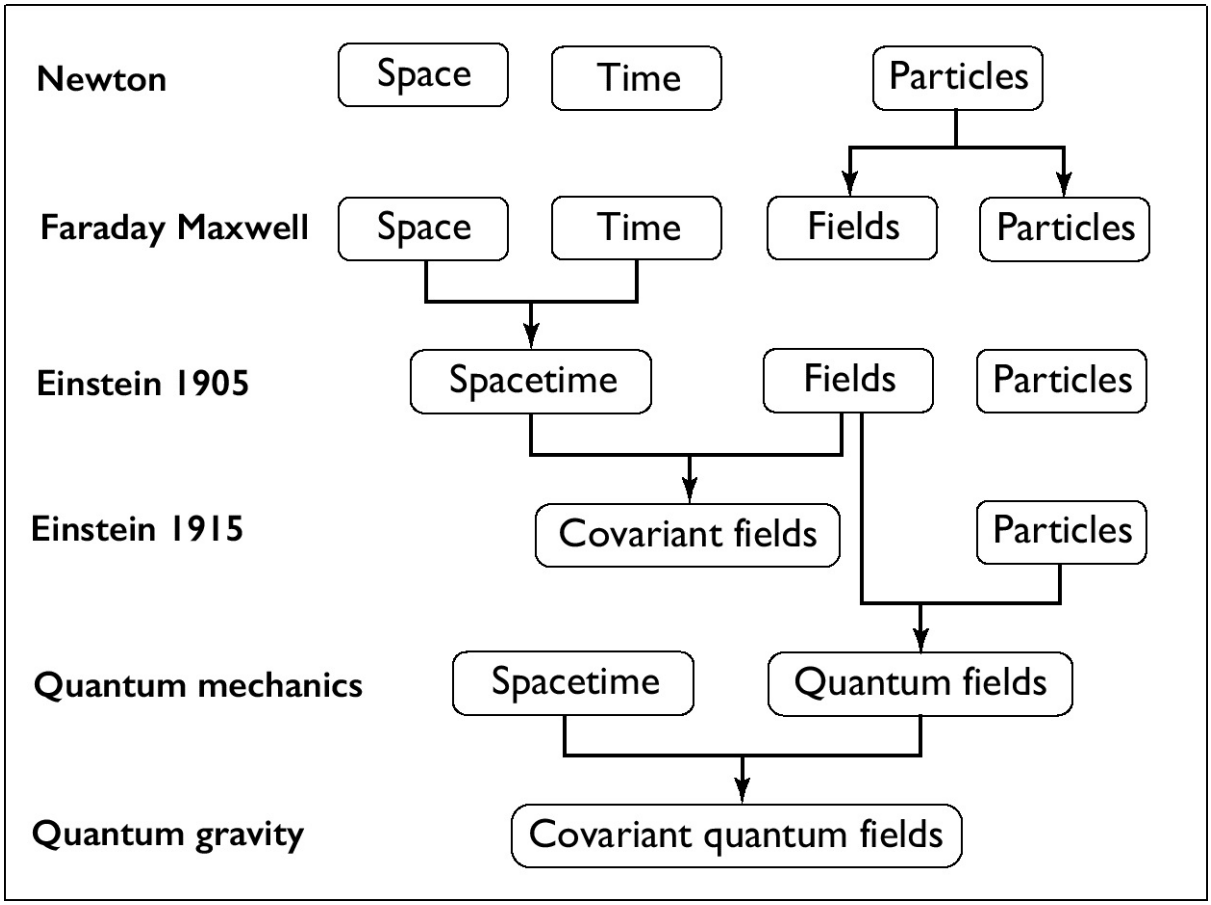


Figure 7.8 What is the world made of? Of only one ingredient: covariant quantum fields.

Covariant quantum fields have become today the best description that we have of the ἄπειρον, the *apeiron*, the primal substance of which everything is formed hypothesized by the man that could perhaps be called the first scientist and the first philosopher, Anaximander.[fn45]

The separation between the curved and continuous space of Einstein's general relativity and the discrete quanta of quantum mechanics which dwell in a flat and uniform space has dissolved. The apparent contradiction is no longer there. Between the spacetime continuum and quanta of space, there is